



Northern Beaches Secondary College

Manly Selective Campus

2014 HSC Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators and templates may be used.
- All necessary working should be shown in every question.
- Multiple choice questions are to be completed on the special answer page.

Total marks – 70

- Attempt Questions 1-14

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Multiple choice section

Answer each of the following ten (10) questions on the special answer sheet provided.

Q1. $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ is equivalent to?

- A) $\cos 2\theta$
- B) $\sin 2\theta$
- C) $\tan 2\theta$
- D) $\cot 2\theta$

Q2. The remainder theorem when $P(x) = x^3 - 2x^2 - 4x + 7$ is divided by $2x + 3$ is:

- A) 4
- B) $\frac{41}{8}$
- C) $-\frac{1}{8}$
- D) -26

Q3. The point $P(2, 2)$ divides the interval joining $A(-2, -4)$ and $B(x_2, y_2)$ in the ratio 2:1. The coordinates of B are?

- A) (4, 5)
- B) (6, 8)
- C) (0, -1)
- D) (10, 14)

Q4. The term independent of x in the expansion $\left(x + \frac{3}{x}\right)^4$ is?

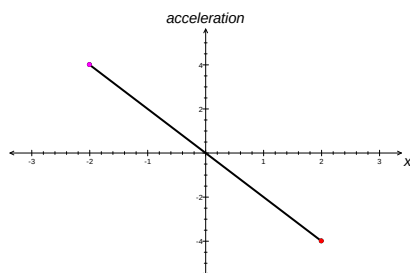
- A) 3
- B) 6
- C) 18
- D) 54

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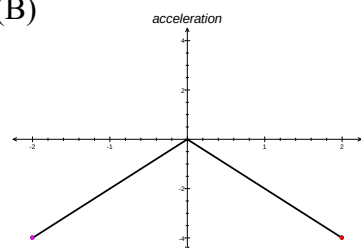
- Q5. A particle is moving along a straight line. The displacement of the particle from a fixed point O is given by x . The graphs below show acceleration against displacement.

Which of the graphs below best represents a particle moving in simple harmonic motion?

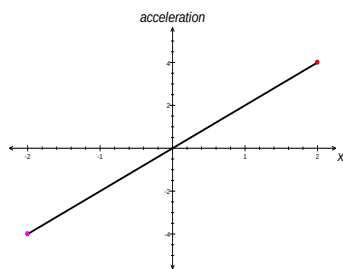
(A)



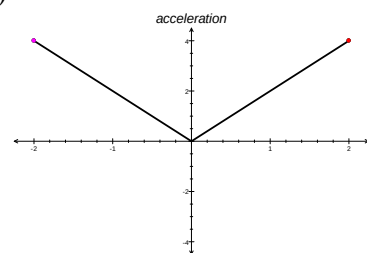
(B)



(C)



(D)



- Q6. The definite integral $\int_e^{e^2} \frac{2}{x(\log_e x)^2} dx$ is evaluated using the substitution $u = \log_e x$.

The value of the integral is?

A) $2\left(\frac{1}{e} - \frac{1}{e^2}\right)$

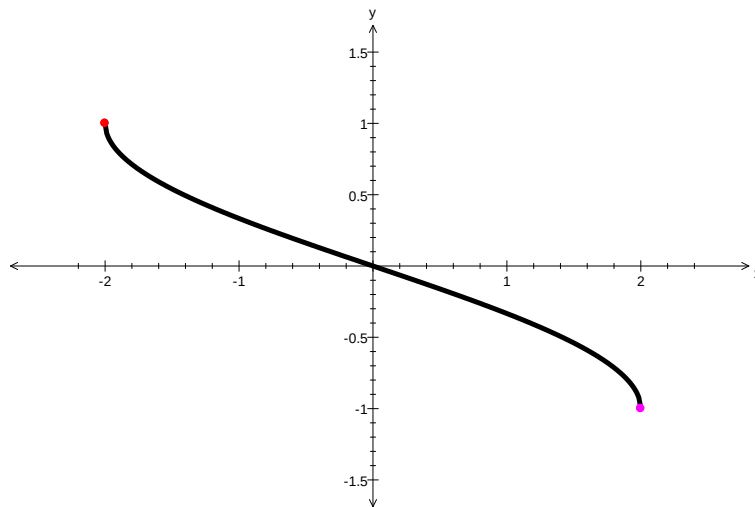
B) $2\left(\frac{1}{e^2} - \frac{1}{e}\right)$

C) -1

D) 1

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- Q7. The function shown in the diagram below has the equation $y = A \sin^{-1} Bx$. Which of the following is true?

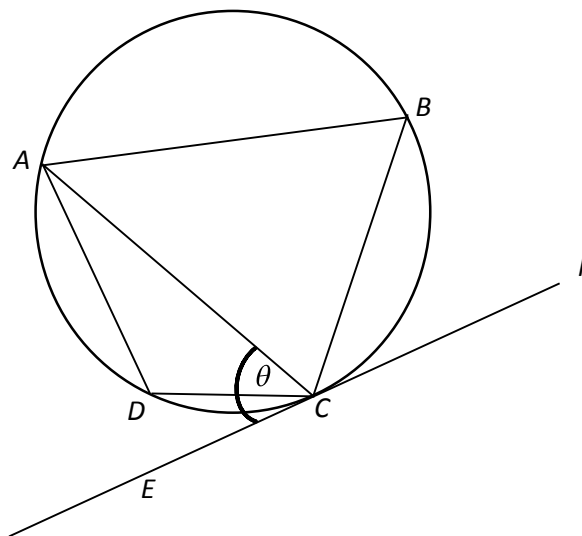


- (A) $A = 1, B = \frac{1}{2}$
- (B) $A = -1, B = 2$
- (C) $A = \frac{-2}{\pi}, B = \frac{1}{2}$
- (D) $A = \frac{2}{\pi}, B = 2$
- Q8. Three English books, four Mathematics books and five Science books are randomly placed along a bookshelf. What is the probability that the Mathematics books are all next to each other?

- A) $\frac{1}{3!5!}$
- B) $\frac{4!9!}{12!}$
- C) $\frac{4!3!5!}{12!}$
- D) $\frac{4!}{9!}$

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- Q9. A, B, C and D are points on a circle. The line l is tangent to the circle at C .
 $\angle ACE = \theta$.



What is $\angle ADC$ in terms of θ ?

- A) $90^\circ - \theta$
 - B) $180^\circ - \theta$
 - C) $180^\circ - 2\theta$
 - D) θ
- Q10. What is the indefinite integral for $\int (\cos^2 x + \sec^2 x) dx$?

- A) $\frac{1}{2}x + \frac{1}{4}\sin 2x + \frac{1}{2}\tan x + c$
- B) $\frac{1}{2}x - \frac{1}{4}\sin 2x + \frac{1}{2}\tan x + c$
- C) $\frac{1}{2}x + \frac{1}{4}\sin 2x + \tan x + c$
- D) $\frac{1}{2}x - \frac{1}{4}\sin 2x + \tan x + c$

End of Multiple Choice

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Free response questions – answer each question in a separate Booklet

Question 11: Start a new Booklet

15 Marks

a) Solve $\left(\frac{x+3}{x^2-1}\right) \leq 0$ 2

- b) A container ship brings a total of 1200 cars into Australia. Of these cars, three hundred have defective brakes.

A total of five hundred cars are unloaded at Sydney.

Find an expression for the probability that one hundred of the cars unloaded at Sydney have defective brakes. 2

(NOTE: You don't need to simplify or evaluate the expression).

- c) How many times must a die be rolled so that the probability of at least one six exceeds 0.5? 2

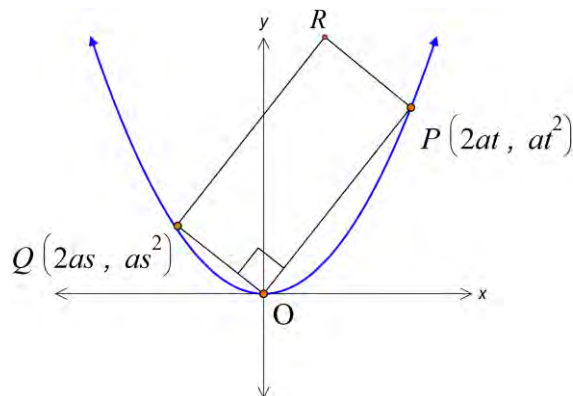
- d) Line A is defined by the equation $y = 2x + 1$.
Line B is defined by the equation $y = mx + b$

If the acute angle between the two is 45 degrees, what are the possible values of m ? 2

Question 11: continued on next page

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Question 11: continued.



- e) In the diagram above the point R is the fourth vertex of the rectangle $POQR$.
 The points P and Q move such that $\angle POQ = 90^\circ$.

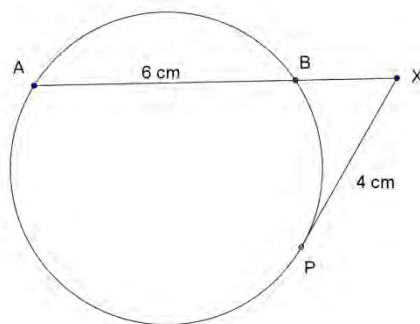
(i) Show that $st = -4$ 1

(ii) Find the locus of the point R 2

- f) Find the general solution of the equation: 2

$$\sin\theta \cos\theta = \frac{1}{2}$$

- g) The diagram shown below shows a circle and AB is a chord of length 6 cm.
 The tangent to the circle at P meets AB produced at X . $PX = 4$ cm.
 Find the length of BX . 2



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Question 12: Start a new Booklet

15marks

- a) (i) Show that a zero of the function

$$f(x) = \log_e x - \frac{1}{x}$$

lies between $x = 1$ and $x = 2$.

1

- (ii) Use one application of Newton's Method with an initial approximation of $x = 1.5$ to obtain an improved estimate to the solution of the equation:

$$\log_e x - \frac{1}{x} = 0.$$

(State your answer to one decimal place.)

2

- b) The rate of change of the temperature T of a cool item placed in a hot environment is determined by the equation.

$$\frac{dT}{dt} = k(S - T)$$

where k is a constant and T is the temperature of the object, and S is the temperature of the environment.

- (i) Show that $T = S - Ae^{-kt}$ is a solution to the differential equation:

$$\frac{dT}{dt} = k(S - T).$$

1

Jamie is cooking a large roast in an oven set to 160°C . The roast will be cooked when the thermometer shows that the temperature of the centre of the roast is 150°C . When Jamie started cooking, the temperature of the centre of the roast was 4°C and 30 minutes later it was 60°C .

- (ii) How long will it take for the roast to be cooked?

3

Question 12 continued on next page

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Question 12 continued

c) (i) Show that $\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{1}{(n+2)!}$ **1**

(ii) Use mathematical induction to show that, for all integers $n \geq 1$,

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!} .$$
3

d) Given the function $f(x) = \frac{x^2 - 1}{x^3 - 8}$

(i) State the vertical asymptote. **1**

(ii) Sketch the graph of the function. Clearly show on your diagram the x – intercepts. **3**

(Your diagram should be one third of a page in size)

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Question 13: Start a new Booklet

15 marks

a) Find $\int e^{x+e^x} dx$ using the substitution $u = e^x$ **3**

- b) A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line. The velocity is given by

$$\dot{x} = -\frac{1}{8}x^3$$

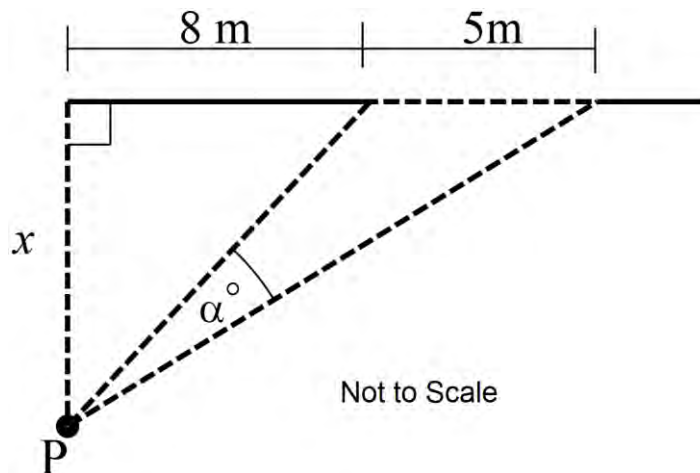
The acceleration of the particle is given by $\ddot{x}$. The particle is initially 2 metres to the right of O .

- (i) Show that $\ddot{x} = \frac{3}{64}x^5$ **2**
- (ii) Find an expression for x in terms of t **3**
- c) The letters of the word **INTEGRAL** are arranged in a row. Calculate the probability that there are three letters between the letters “N” and “T”. **3**

Question 13 continued on next page

Question 13 continued

- d) During the medieval wars, the enemy wanted to attack a fortress with a 5 metre opening along a front wall. The strategy was to stand at the point P, on a line 8 metres from the opening and perpendicular to the wall, as per the diagram. The archer stands x metres away from the wall, thus giving an angle of vision, α , through which to fire arrows from a cross-bow.



- (i) Show that the angle of vision α is given by

$$\alpha = \tan^{-1}\left(\frac{13}{x}\right) - \tan^{-1}\left(\frac{8}{x}\right) \quad 1$$

- (ii) Determine the distance x which gives the maximum angle of vision α . 3

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Question 14: Start a new Booklet

15 marks

a) Let α, β, γ be the roots of $P(x) = 2x^3 - 5x^2 + 3x - 5$.

Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

3

b) Let $(3 + 2x)^{20} = \sum_{r=0}^{20} a_r x^r$

(i) Write down an expression for a_r

1

(ii) Show that $\frac{a_{r+1}}{a_r} = \frac{40-2r}{3r+3}$

2

(iii) Find the value of r which produces the greatest coefficient in the expansion of $(3 + 2x)^{20}$

1

c) A particle is moving in a straight line in simple harmonic motion. At time t it has displacement x metres from a fixed point O on the line where

$$x = (\cos t + \sin t)^2$$

At time t , the velocity of the particle is $\dot{x} \text{ ms}^{-1}$ and the acceleration is $\ddot{x} \text{ ms}^{-2}$

(i) Show that $\ddot{x} = -4(x - 1)$

2

(ii) Find the extreme positions of the particle during its motion.

2

Question 14 continued on next page.

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Question 14 continued

- d) (i) Show that:

$$(1-x)^n \left(1 + \frac{1}{x}\right)^n = \left(\frac{1-x^2}{x}\right)^n \quad 1$$

- (ii) By considering the expansion of $(1-x)^n \left(1 + \frac{1}{x}\right)^n$ or otherwise, express

$$\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2}$$

in simplest form.

3

END OF EXAMINATION PAPER

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

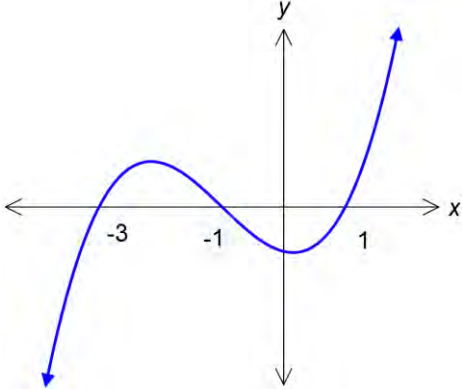
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1	Let $t = \tan \theta$ then $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2t}{1+t^2}$ $= \sin 2\theta$	B
Q2	$x = -\frac{3}{2}$ $\left(-\frac{3}{2}\right)^3 - 2 \times \left(-\frac{3}{2}\right)^2 - 4 \times \left(-\frac{3}{2}\right) + 7 = \frac{41}{8}$	B
Q3	$\frac{1(-2)+2x}{2+1} = 2 \Rightarrow x = 4$ $\frac{1(-4)+2y}{2+1} = 2 \Rightarrow y = 5$	A
Q4	$T_{k+1} = {}^4C_k x^{4-k} \left(\frac{3}{x}\right)^k$ $= {}^4C_k x^{4-k} (3^k x^{-k})$ $= {}^4C_k 3^k x^{4-2k}$ independent of $x \Rightarrow 4 - 2k = 0$ $k = 2$ the term is ${}^4C_2 \times 3^2 = 54$	D
Q5	For simple harmonic motion $\ddot{x} = -n^2 x$	A

Q6	$\int_e^{e^2} \frac{2}{x(\log_e x)^2} dx = 2 \int_e^{e^2} \frac{1}{(\log_e x)^2} \times \frac{1}{x} dx$ $= 2 \int_1^2 \frac{1}{u^2} du$ $= 2 \int_1^2 u^{-2} du$ $= 2 \left[\frac{u^{-1}}{-1} \right]_1^2$ $= -2 \left[\frac{1}{u} \right]_1^2$ $= -2 \left(\frac{1}{2} - 1 \right)$ $= 1$	D
Q7	$A = \frac{-2}{\pi}$ Domain is $D: -1 \leq Bx \leq 1$ $-\frac{1}{B} \leq x \leq \frac{1}{B}$ $\therefore \frac{1}{B} = 2$ $B = \frac{1}{2}$	C
Q8	$\frac{4!9!}{12!}$	B
Q9	$180^\circ - \theta$	B
Q10	$\frac{1}{2}x + \frac{1}{4}\sin 2x + \tan x + c$	C

Q11

Q11 -a	$\left(\frac{x+3}{x^2-1}\right) \leq 0 \quad nb \ x \neq \pm 1$ $\left(\frac{x+3}{x^2-1}\right)(x^2-1)^2 \leq 0 \times (x^2-1)^2$ $(x+3)(x^2-1) \leq 0$ $(x+3)(x-1)(x+1) \leq 0$  therefore $x \leq -3$ or $-1 < x < 1$	2 marks: correct solution 1 mark: $x \neq 1$ or -1 and $(x+3)(x^2-1) \leq 0$
Q11 -b	$P_{\text{defective}} = p = \frac{1}{4}$ $P_{\text{non-defective}} = q = \frac{3}{4}$ $(p+q)^{500}$ $100 \text{ defective} = {}^{500}\text{C}_{400} p^{100} q^{400}$ $= {}^{500}\text{C}_{400} \left(\frac{1}{4}\right)^{100} \left(\frac{3}{4}\right)^{400}$	2 marks: correct solution 1 mark: $p = \frac{1}{4}, q = \frac{3}{4}$ and considering $(p+q)^{500}$

Q11-c	$p_6 = \frac{1}{6} \quad q_{not\ 6} = \frac{5}{6}$ $\therefore p_{no\ 6's} = \left(\frac{5}{6}\right)^n$ $\therefore 1 - \left(\frac{5}{6}\right)^n > 0.5$ $0.5 > \left(\frac{5}{6}\right)^n$ $\ln(0.5) > n \ln\left(\frac{5}{6}\right)$ $\frac{\ln(0.5)}{\ln\left(\frac{5}{6}\right)} < n$ $3.8 < n$ $\therefore n = 4$ Therefore 4 throws of dice required.	2 marks: correct solution 1 mark: $1 - \left(\frac{5}{6}\right)^n > 0.5$ Or bald correct answer Note: many students failed to recognise that $\ln\left(\frac{5}{6}\right)$ is negative requiring inequality sign to be reversed
Q11-d	$y = 2x + 1 \quad m_1 = 2 \quad m_2 = m$ $\tan\theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $1 = \left \frac{2 - m}{1 + 2m} \right $ $-1 = \frac{2 - m}{1 + 2m} \quad 1 = \frac{2 - m}{1 + 2m}$ $-2m - 1 = 2 - m \quad 1 + 2m = 2 - m$ $m = -3 \quad m = \frac{1}{3}$	2 marks: correct solution 1 mark: correct formula

Q11-e-i	$st = -4$ $m_{OQ} = \frac{as^2}{2as} = \frac{s}{2}$ $m_{OP} = \frac{at^2}{2at} = \frac{t}{2}$ $m_{OQ} \times m_{OP} = -1$ $\frac{s}{2} \times \frac{t}{2} = -1$ $\therefore st = -4$	1 mark: correct demonstration
Q11-e-ii	$Q(2as, as^2) \quad P(2at, at^2)$ <i>As shape is a rectangle</i> $R = (2as + 2at, as^2 + at^2)$ $x = 2a(s + t) \quad y = a(s^2 + t^2)$ $\frac{x}{2a} = (s + t)$ $(s + t)^2 = s^2 + 2st + t^2$ $(s + t)^2 - 2st = s^2 + t^2$ $\frac{x^2}{4a^2} - 2(-4) = s^2 + t^2$ $\therefore y = a\left(\frac{x^2}{4a^2} + 8\right)$ $y = \frac{x^2}{4a} + 8a$ $4ay = x^2 + 32a^2$ $x^2 = 4a(y - 8a)$	2 marks: correct demonstration 1 mark: applying $(s + t)^2 = s^2 + 2st + t^2$ Note: too many students were unable to write down coords of R, but wasted time deriving them
Q11-f	$\sin\theta \cos\theta = \frac{1}{2}$ $\therefore 2\sin\theta\cos\theta = 1$ $\sin 2\theta = 1$ $2\theta = \frac{\pi}{2} + 2n\pi$ $\theta = \frac{\pi}{4} + n\pi$	2 marks :correct solution 1 mark: correct base value

Q11-g	$\text{XB. } AX = XP^2$ $(6 + x)x = 16$ $x^2 + 6x - 16 = 0$ $(x + 8)(x - 2) = 0$ $x = -8 \text{ or } 2$ $\therefore x = 2 \text{ distance}$	2 marks: correct solution 1 mark: correct equation $(6 + x)x = 16$
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Q12

Q12-a-i	$f(x) = \log_e x - \frac{1}{x}$ $f(1) = \log_e 1 - \frac{1}{1} = 0 - 1 = \text{neg}$ $f(2) = \log_e 2 - \frac{1}{2} = 0.19 = \text{pos}$ Therefore at least one root must lie between 1 and 2 as graph moves from negative to positive.	1 mark – correct explanation based on correct values calculated.
Q12-a-i	$f(x) = \log_e x - \frac{1}{x}$ $f'(x) = \frac{1}{x} + \frac{1}{x^2} = \frac{x+1}{x^2}$ $a_2 = a_1 - \frac{f(x)}{f'(x)}$ $= 1.5 - \frac{\log_e 1.5 - \frac{1}{1.5}}{\frac{1.5+1}{(1.5)^2}}$ $= 1.735$	2 marks – correct solution 1 mark – correct substitution into correct formula
Q12b-i	Version 1 $T = S - Ae^{-kt}$ $\frac{dT}{dt} = kAe^{-kt}$ $= kS - kS + kAe^{-kt}$ $= -k(-S + S - Ae^{-kt})$ $= -k(-S + T)$ $= k(S - T)$ Version 2 $T = S - Ae^{-kt}$ $\therefore Ae^{-kt} = S - T \quad \text{①}$ $\frac{dT}{dt} = kAe^{-kt}$ $= k(S - T) \quad \text{from ①}$	1 mark – fully explained in explanation of substitution.

q12b-ii	$T = S - Ae^{-kt}$ at $t = 0$ $T = 4$ $4 = 160 - Ae^0$ $A = 156$ at $t = 30$ $T = 60$ $60 = 160 - 156e^{-30k} \quad \textcircled{1}$ $e^{-30k} = \frac{100}{156}$ $-30k = \ln \frac{100}{156}$ $k = \ln \frac{100}{156} \div -30 = 0.014823$ $150 = 160 - 156e^{-kt}$ $e^{-kt} = \frac{10}{156}$ $t = \ln \frac{10}{156} \div -k$ $= \ln \frac{10}{156} \div \left(-\ln \frac{100}{156} \div -30 \right)$ $= 185 \text{min } 20 \text{sec}$	3 marks – correct solution 2 marks – correct value for k 1 mark – correct to line (1)
Q12-c	$\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!}$ $= \frac{1}{(n+1)!} - \frac{n+1}{(n+2)(n+1)!}$ $= \frac{n+2}{(n+2)(n+1)!} - \frac{n+1}{(n+2)!}$ $= \frac{(n+2) - (n+1)}{(n+2)!}$ $= \frac{1}{(n+2)!}$	1 mark – correct solution

Q12c-ii	$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$ For $n = 1$ $LHS = \frac{1}{(1+1)!} = \frac{1}{2}$ $RHS = 1 - \frac{1}{(1+1)!} = \frac{1}{2}$	
Q12c-ii	$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$ Assume true for $n = k$ ie. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$ RTP true for $n = k+1$ $LHS = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{((k+1)+1)!}$ $= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!}$ $= 1 - \left\{ \frac{1}{(k+1)!} - \frac{k+1}{(k+2)!} \right\}$ $= 1 - \frac{1}{(k+2)!}$ $= RHS$ Therefore proved true for all n by process of mathematical induction.	3 marks – correct solution clearly showing use of assumption and or substitution or matching algebra 2 marks – solution partially showing use of assumption and or substitution or matching algebra 1 mark – correct for $n = 1$
Q12d-i	$f(x) = \frac{x^2 - 1}{x^3 - 8}$ $= \frac{x^2 - 1}{(x-2)(x^2 + x + 4)}$ $\therefore x = 2$ is vertical asymptote	1 mark – correct solution $x \neq 2$ not accepted.
Q12d-ii		3 marks - correct shape - correct asymptotes - correct x intercepts

Q13

Q13-a	$\int e^{x+e^x} dx$ $u = e^x \quad du = e^x dx$ $= \int e^x \times e^{e^x} dx$ $= \int e^u du$ $= e^u + C$ $= e^{e^x} + C$	3 marks – correct solution 2 marks – for correct expression $\int e^u du = e^u + C$ 1 mark for $e^{x+e^x} = e^x \cdot e^{e^x}$
Q13b-i	$\dot{x} = -\frac{1}{8}x^3 = v$ $\therefore \frac{1}{2}v^2 = -\frac{1}{128}x^6$ $\ddot{x} = \frac{d\left(\frac{1}{2}v^2\right)}{dx} = \frac{6}{128}x^5$ $= \frac{3}{64}x^5$	2 marks for correct solution. 1 mark for $\frac{1}{2}v^2 = -\frac{1}{128}x^6$

Q13b-ii	$\dot{x} = -\frac{1}{8}x^3$ $\frac{dx}{dt} = -\frac{1}{8}x^3$ $\therefore \frac{dt}{dx} = -\frac{8}{x^3} = -8x^{-3}$ $t = \int -8x^{-3} dx = \frac{-8x^{-2}}{-2}$ $t = \frac{4}{x^2} + C$ at $t = 0 \qquad x = 2$ $0 = \frac{4}{4} + C$ $\therefore C = -1$ $t = \frac{4}{x^2} - 1$ $t + 1 = \frac{4}{x^2}$ $x^2 = \frac{4}{t + 1}$ $x = \sqrt{\frac{4}{t + 1}}$ Note; Positive answer only to agree original conditions	3 marks for correct solution 2 marks - $t = \frac{4}{x^2} + C$ and $C = -1$ - $t = \frac{4}{x^2} + C$ and correct primitive from incorrect value for C 1 mark - $t = \frac{4}{x^2}$																																
Q13c	INTEGRAL <table border="1"><tr><td>N</td><td></td><td></td><td></td><td>T</td><td></td><td></td><td></td></tr><tr><td></td><td>N</td><td></td><td></td><td></td><td>T</td><td></td><td></td></tr><tr><td></td><td></td><td>N</td><td></td><td></td><td></td><td>T</td><td></td></tr><tr><td></td><td></td><td></td><td>N</td><td></td><td></td><td></td><td>T</td></tr></table> <i>Total Number of arrangements</i>= $8!$ <i>Number of restricted arrangements</i>= $4 \times 2! \times 6!$ $Prob = \frac{4 \times 2! \times 6!}{8!} = \frac{1}{7}$	N				T					N				T					N				T					N				T	3 marks – correct solution 2 marks – for $n(S)=8!$ and any two correct of 4 or 2! or 6! 1 mark - $8!$ -any two of 4 or 2! or 6!
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Q13-d	$\alpha = \tan^{-1}\left(\frac{13}{x}\right) - \tan^{-1}\left(\frac{8}{x}\right)$ $\alpha = \angle QPS - \angle QPR$ $\tan(\angle QPR) = \frac{8}{x} \quad \tan(\angle QPS) = \frac{13}{x}$ $\angle QPR = \tan^{-1}\frac{8}{x} \quad \angle QPS = \tan^{-1}\frac{13}{x}$ $\therefore \alpha = \tan^{-1}\left(\frac{13}{x}\right) - \tan^{-1}\left(\frac{8}{x}\right)$	1 mark for correct solution
Q13-d-ii	$\alpha = \tan^{-1}\left(\frac{13}{x}\right) - \tan^{-1}\left(\frac{8}{x}\right)$ $\frac{d\alpha}{dx} = -\frac{13}{x^2}\left(\frac{1}{1 + \left(\frac{13}{x}\right)^2}\right) + \frac{8}{x^2}\left(\frac{1}{1 + \left(\frac{8}{x}\right)^2}\right)$ $\therefore \text{at } \frac{d\alpha}{dx} = 0$ $\frac{13}{x^2}\left(\frac{1}{1 + \left(\frac{13}{x}\right)^2}\right) = \frac{8}{x^2}\left(\frac{1}{1 + \left(\frac{8}{x}\right)^2}\right)$ $\frac{8x^2}{x^2 + 64} = \frac{13x^2}{x^2 + 169}$ $8(x^2 + 169) = 13(x^2 + 64)$ $x^2 = \frac{520}{5}$ $x = \sqrt{104}$ Test for maximum $x = 10 \quad \frac{d\alpha}{dx} = -\frac{13}{100}\left(\frac{1}{1 + \left(\frac{13}{10}\right)^2}\right) + \frac{8}{100}\left(\frac{1}{1 + \left(\frac{8}{10}\right)^2}\right) = 0.0004$ $x = 10.2 \quad \frac{d\alpha}{dx} = -0.000004$ Therefore change in gradient positive, zero to negative therefore maximum.	3 marks – correct solution 2 marks - $x = \sqrt{104}$ 1 mark - correct for $\frac{d\alpha}{dx}$ - value of x correctly obtained from incorrect $\frac{d\alpha}{dx}$

Q14-a	$\alpha + \beta + \gamma = -\frac{b}{a} = \frac{5}{2}$ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{3}{2}$ $\alpha\beta\gamma = -\frac{d}{a} = \frac{5}{2}$ $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2}$ $(\alpha\beta + \alpha\gamma + \beta\gamma)^2$ $= \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 + 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$ $= \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)$ $\therefore \left(\frac{3}{2}\right)^2 = \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 + 2 \times \frac{5}{2} \times \frac{5}{2}$ $\frac{9}{4} - \frac{25}{2} = \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$ $\frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2} = \left(-\frac{41}{4}\right) \div \frac{25}{4} = -\frac{41}{25}$	3 marks: correct solution 2 marks : partial correct with a least 2 values of ratios of coefficients correct with correct required expansion 1 mark: at least two ratios correct only Or 1 mark: correct required expansion only
	$(3 + 2x)^{20}$ $\therefore x^r \text{ term} = {}^{20}\text{C}_r 3^{20-r} \cdot (2x)^r$ $a_r = {}^{20}\text{C}_r 3^{20-r} \cdot (2)^r$	1 mark: correct solution

	$a_r = {}^{20}C_r 3^{20-r} \cdot (2)^r$ $a_{r+1} = {}^{20}C_{r+1} 3^{19-r} \cdot (2)^{r+1}$ $\frac{a_{r+1}}{a_r} = \frac{{}^{20}C_{r+1} 3^{19-r} \cdot (2)^{r+1}}{{}^{20}C_r 3^{20-r} \cdot (2)^r}$ $= \frac{20! \times 3^{19-r} \cdot (2)^{r+1}}{(19-r)!(r+1)!} \times \frac{(20-r)!r!}{20! \times 3^{20-r} \cdot (2)^r}$ $= \frac{2(20-r)}{3(r+1)}$ $= \frac{40-2r}{3r+3}$	2 marks: correct solution 1 mark: partial correct with both terms correct
	$\frac{40-2r}{3r+3} \leq 1$ $40-2r \leq 3r+3$ $37 \leq 5r$ $7.4 \leq r$ $\therefore r = 8$ $r = 8 \quad {}^{20}C_8 3^{12} \times 2^8 = 1.714 \times 10^{13}$ $r = 7 \quad {}^{20}C_7 3^{13} \times 2^7 = 1.582 \times 10^{13}$ $\text{Test } r = 9 \quad {}^{20}C_9 3^{11} \times 2^9 = 1.523 \times 10^{13}$	1 mark: correct solution
14c-i	$x = (\cos t + \sin t)^2 = \cos^2 t + 2\sin t \cos t + \sin^2 t$ $\therefore x = 1 + \sin 2t$ $\dot{x} = 2\cos 2t$ $\ddot{x} = -4\sin 2t$ $= -4(1 + \sin 2t - 1)$ $= -4(x - 1)$	2 marks: correct solution 1 mark: correct expression for velocity
14c-ii	$x = 1 + \sin 2t$ $-1 \leq \sin 2t \leq 1$ $0 \leq 1 + \sin 2t \leq 2$ $0 \leq x \leq 2$ therefore extremes of position are $x = 0$ and $x = 2$	2 marks: correct solution 1 mark: only one correct extreme

14 d -i	$\begin{aligned}(1-x)^n\left(1+\frac{1}{x}\right)^n &= \left[(1-x)\left(1+\frac{1}{x}\right)\right]^n \\&= \left[1+\frac{1}{x}-x-1\right]^n \\&= \left(\frac{1}{x}-x\right)^n \\&= \left[\frac{1-x^2}{x}\right]^n\end{aligned}$	1 mark: correct solution
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14d-ii	Determine the coefficient of x^2 on both sides of the equation. $RHS = (1-x)^n \left(1 + \frac{1}{x}\right)^n$ $= \left[\sum_{k=0}^n (-1)^k {}^nC_k x^k \right] \left[\sum_{k=0}^n {}^nC_k x^{-k} \right]$ $\therefore$ coefficient of x^2 $\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2}$ $LHS = \left(\frac{1-x^2}{x}\right)^n = \left(\frac{1}{x} - x\right)^n$ $= \sum_{k=0}^n (-1)^k \binom{n}{k} x^{-(n-k)} x^k$ $= \sum_{k=0}^n (-1)^k \binom{n}{k} x^{2k-n}$ $\therefore$ coefficient of x^2 at $2k - n = 2$ $k = 1 + \frac{n}{2}$ If n is odd, then k is not an integer therefore cannot exist. therefore $\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2} = 0$ If n is even then $(-1)^{1+\frac{n}{2}} \binom{n}{1+\frac{n}{2}}$ is coefficient of x^2 therefore $\binom{n}{2}\binom{n}{0} - \binom{n}{3}\binom{n}{1} + \dots + (-1)^n \binom{n}{n}\binom{n}{n-2} = (-1)^{1+\frac{n}{2}} \binom{n}{1+\frac{n}{2}}$	3 marks : correct solution 2 marks: partial correct with significant progress to equating coefficients of squared term 1 mark: one expression for the squared term correct
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